The Keystone: From $(\mathbb{S}, \tau)$ to the Injective Type III$_1$ Factor

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A standalone, sequential compilation of the construction Foundation cites by title. Every proposition below is proved, verified, or cited with a locator, in that order of preference; nothing is asserted on authority alone. Compiled from ten source papers of the wider archive (listed in full in §14), reorganized into one continuous argument — plus one section, §9, derived within this project's own review correspondence rather than drawn from the archive: the chain's first original extension, not a compilation of prior work. One gap this compilation closed with a standard argument; one it located for the first time by putting the chain in one place, and which a companion paper has since resolved as a proven theorem — the chain's second original extension, cited here rather than reproduced; one it carries as a declared conditionality worth more than a footnote.

Revision 19. Closes the locator debt in §3's X/P identification item, checked directly against Günaydin–Lüst–Malek's body equations rather than the abstract alone. Their string-level contraction (§3.2 of that paper, presented as a direct demonstration, not hedged as a conjecture) gives an explicit generator correspondence: momenta $\propto$ the closed, associative quaternion triple $\{e_1, e_2, e_3\}$; coordinates $\propto$ the non-closed triple $\{e_4, e_5, e_6\}$, which after contraction carries the surviving nonzero associator. This is GLM's own explicit, named identification — coordinates as the non-associative triple — not an artifact of a commutator sign, and stronger evidence for Orbit B's identification than the previous convention-level check, though still not a derivation internal to this document's own construction. The physical question itself remains open; the evidentiary weight on one side of it has changed.

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1. The split

[DEF] $\mathbb{S}$ is the sixteen-dimensional real Cayley–Dickson algebra obtained from one further doubling of the octonions $\mathbb{O}$: $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$, unital, flexible, quadratic, power-associative, with positive-definite norm, and — with the same weight as its virtues — not associative, not alternative, possessed of zero divisors.

[FACT, proven] $\mathbb{O}e_8$ fails to close under its own product: for $m, n \in \mathbb{O}e_8$, $mn \in \mathbb{O}$, not $\mathbb{O}e_8$. This non-closure pattern — a doubling's second half self-multiplying back into the first — is not itself unique to $\mathbb{S}$; it recurs at every level of the tower (already at $\mathbb{C} = \mathbb{R} \oplus \mathbb{R}i$, $i \cdot i = -1 \in \mathbb{R}$). What is unique to $\mathbb{S}$ is sharper: it is the *last* level at which the representable half retains genuinely non-Lie Malcev structure — rich enough for the enveloping-algebra machinery below — and the construction cannot be repeated one level further ($\mathbb{S} \oplus \mathbb{S}e_{16}$), since $\mathbb{S}$'s own commutator fails the Malcev identity outright, so $\mathbb{S}$ cannot play the role $\mathbb{O}$ plays here.

[DEF] τ is the involution fixing $\mathbb{O}$ pointwise and negating $\mathbb{O}e_8$: $\mathbb{O} = \mathrm{Fix}(\tau)$, $\mathbb{O}e_8 = \ker(\tau + 1)$. $\mathrm{Aut}(\mathbb{S}) = G_2 \times S_3$ is classical (Eakin–Sathaye, 1990); the S_3 factor permutes three conjugate choices of such a split, related by an order-three element μ.

[FACT, proven, addressing a literature discrepancy] A recent preprint (Wilmot, arXiv:2512.07210) disputes this count, arguing for Schafer's $\mathrm{Aut}(\mathbb{S}) \cong G_2$ alone. That claim is refuted in three steps, each independently checkable. *(i)* τ is an automorphism, by direct computation from the doubling formula $(a+be_8)(c+de_8) = (ac-\bar{d}b)+(da+b\bar{c})e_8$: applying τ to the product gives $(ac-\bar{d}b)-(da+b\bar{c})e_8$, while $\tau(a+be_8)\cdot\tau(c+de_8) = (a-be_8)(c-de_8) = (ac-(-\bar{d})(-b))+((-d)a+(-b)\bar{c})e_8$ — the same element, for all $a, b, c, d \in \mathbb{O}$, by bilinearity alone; nothing octonion-specific is used, so the analogous involution exists at every Cayley–Dickson level. (Confirmed numerically: zero defect, 500 random pairs.) *(ii)* τ lies outside the diagonal G_2: any $g(a+be_8) = g(a)+g(b)e_8$ matching τ on $\mathbb{O}$ fixes $\mathbb{O}$ pointwise, hence $g = \mathrm{id}$, which acts as $+1$ on $\mathbb{O}e_8$ where τ acts as -1. *(iii)* The diagonal G_2 is the whole identity component, so *(ii)* refutes the isomorphism claim itself, not merely one embedding of it: $\mathrm{Aut}(\mathbb{S})$ is a compact Lie group (automorphisms preserve the quadratic algebra's trace and norm forms, hence sit in $O(16)$, where they form a closed subgroup since multiplicativity is a closed condition) with Lie algebra $\mathrm{Der}(\mathbb{S})$; solving the derivation equations directly gives $\dim \mathrm{Der}(\mathbb{S}) = 14$ with every solution block-diagonal with equal blocks — exactly the diagonal $\mathfrak{g}_2$, verified to machine precision with an unambiguous spectral gap, and agreeing with the classical derivation count on which both sides of the dispute concur [locator owed: the derivation half of Eakin–Sathaye 1990]. A compact connected group with Lie algebra $\mathfrak{g}_2$ is G_2 itself, so the identity component is the diagonal G_2, τ inhabits a second component, and $\mathrm{Aut}(\mathbb{S})$ is disconnected — not isomorphic to the connected G_2 under any identification. What this establishes, and no more: the G_2-alone count is wrong. It does not re-derive the full $G_2 \times S_3$; the complete count — in particular the order-three μ, which the three-conjugate-splits structure below actually uses — continues to rest on Eakin–Sathaye (1990).

Everything from here on works inside one fixed choice, $\mathbb{O} = \mathrm{Fix}(\tau)$; which of the three is used is a free, symmetric choice, and this document does not depend on which.

[SCOPE] $\mathbb{O}e_8$'s finer classification — it is a faithful, irreducible Cayley bimodule over $\mathbb{O}$, provably not an alternative bimodule, with the same structure recurring identically across every octonion subalgebra of $\mathbb{S}$ tested — motivates why this split is the physically meaningful one, but is not itself load-bearing for the construction below, which uses only $\mathbb{O} = \mathrm{Fix}(\tau)$ and its multiplication. Not reproduced here; cited by title.

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2. The R-flux Malcev algebra, and W as structural $\hbar$

Everything from here proceeds inside $\mathrm{Im}(\mathbb{O})$, the seven-dimensional imaginary part of $\mathbb{O} = \mathrm{Fix}(\tau)$ — the unique seven-dimensional non-Lie simple Malcev algebra, in its compact real form, under the commutator bracket $[e_i, e_j] = e_ie_j - e_je_i$.

[DEF] A *generalized Inönü–Wigner contraction* of $\mathrm{Im}(\mathbb{O})$ along a $1{+}3{+}3$ grading assigns one imaginary unit to a central generator W (grade 3), three to coordinates X^a (grade 1), three to momenta P_a (grade 2), and rescales $X \to \lambda X$, $P \to \lambda^2 P$, $W \to \lambda^3 W$, keeping only the component of each bracket landing exactly at the sum of its factors' grades as $\lambda \to 0$ (all lower-grade components vanish in the limit; no higher-grade component exists, by validity).

[FACT, proven] (Weimar-Woods validity.) For this grade shape, every bracket automatically satisfies the no-divergence condition except $[X, X]$ (grade sum 2), which must carry no component along W (grade 3) — the one nontrivial constraint. Of the $\binom{7}{1}\binom{6}{3} = 140$ possible assignments, exactly **56** satisfy it, verified by direct, exhaustive computation.

[FACT, proven] These 56 form exactly two orbits of 28 under the independently-verified, order-1344 group of sign-and-permutation automorphisms of this basis's multiplication table — a discrete subgroup of the full 14-dimensional $\mathrm{Aut}(\mathbb{O}) = G_2$, and the group actually relevant to permuting this finite set of 56 assignments, distinguished by which of $\{X, P\}$ is itself one of the octonions' seven closed three-element ("Fano-line") subalgebras:

- **Orbit A** (X closed): $[X, X] \subseteq X$ — zero component at grade 2, so $\{X, X\}$ vanishes on contraction.
- **Orbit B** (P closed): $[P, P] \subseteq P$ — automatically zero at grade 4, which does not exist in this shape, at *no* cost — while $[X, X]$, not being confined to X, is forced into the only grade-2 space available, P, and survives nonzero.

Both orbits satisfy the Malcev identity — verified against 20 random rational combinations per case, not merely basis pairs — but for structurally different reasons: Orbit A satisfies it only because it trivially satisfies the strictly stronger Jacobi identity (the resulting algebra is an ordinary, fully associative Heisenberg algebra, $X, X = P, P = 0$); Orbit B satisfies it as the genuine, non-associative content the construction exists to produce. **Only Orbit B — 28 of the 140 assignments, not 56 — gives actual R-flux.** The condition is mechanistic, not coincidental: $\{X, X\} \propto P$ *is* the defining non-associative content, while $\{P, P\} = 0$ is the expected, required form regardless of any choice — so momentum-side closure costs nothing, coordinate-side closure destroys the one structure that matters.

[FACT, proven] The orbit count has a direct combinatorial origin. Fix a Fano line P and an external point $w \notin P$ (the *mediator*, W's direction); the remaining three points are X. Every such pair is automatically valid (the three lines through w partition the other six points into a perfect matching — the matching lemma proved directly in the four-level-shape paper, §2, and cited here rather than re-derived — each pair meeting P in exactly one point, so no X-product can reach w), and satisfies the *mediator property* $[w, P] \subseteq X$ (for $p \in P$, the third point of the line through w and p lies in neither P nor $\{w\}$, hence in X). Genuine gradings are exactly such pairs: $7 \times 4 = 28$, exactly Orbit B, verified without exception. In the standard Cayley–Dickson labeling this is exact and sign-free: $e_i e_4 = e_{i+4}$ for $i = 1, 2, 3$.

Fixing this grading (one Fano line as P, one external point e_4 as mediator), the contracted bracket is

$$\{X^a, X^b\} = 2\epsilon^{abc} P_c, \qquad \{X^a, P_b\} = -2\delta^a_b W, \qquad \{P_a, P_b\} = 0, \qquad \{X, W\} = \{P, W\} = 0,$$

with Jacobiator $J(X^1, X^2, X^3) = \pm 12W$: nonzero, genuine non-associativity. Fixing $\{X^a, P_b\} = \kappa\delta^a_b$ forces $W = -\kappa/2$ and the Jacobiator coefficient to exactly 6κ — the canonical pairing and the non-associative strength are not independent parameters but two faces of one value, W.

[FACT, proven] (W is structurally central, at every degree.) This algebra is a genuine Malcev algebra, so its universal nonassociative enveloping algebra $U(M) = F(M)/R(M)$ rigorously exists (Pérez-Izquierdo–Shestakov, 2004), and the canonical map $M \to U(M)$ lands inside the generalized alternative nucleus $N_{\mathrm{alt}}(A) = \{a : (a, x, y) = -(x, a, y) = (x, y, a)\ \forall x, y\}$. A direct calculation at degree 2, using this alternation property and $\{X, W\} = \{P, W\} = 0$, gives $[xy, W] = 3(W, x, y)$ for $x, y \in M$, and evaluating the associator explicitly shows this vanishes for every degree-2 combination of the six non-central generators (checked exhaustively, all pairs). The mechanism generalizes: by induction on degree, using the general reduction lemma $[by, a] = [b, a]y + b[y, a] + \frac{1}{2}[[y, a], b] - \frac{1}{2}[[y, b], a] - \frac{1}{2}[y, [a, b]]$ (Pérez-Izquierdo–Shestakov; Bremner–Hentzel–Peresi–Tvalavadze–Usefi), every term reduces to strictly lower degree, closing the induction cleanly. $[x, W] = 0$ **for every basis monomial of $U(M)$, at every degree** — a complete proof, independently confirmed computationally through degree 5 (all $6^2, 6^3, 6^4, 6^5$ generator combinations, zero violations).

By Schur's lemma, a central element of $U(M)$ acts as a fixed scalar in any irreducible representation — the precise structural reason $\hbar$ is a fixed universal constant rather than representation-dependent. W occupies exactly this role, proven rather than assumed by resemblance.

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3. The obstruction, and the second contraction

[FACT, proven] No associative representation of this algebra can carry $W \neq 0$. If $\rho : M \to A$ is linear into an associative algebra with $\rho(\{x, y\}) = [\rho(x), \rho(y)]$, then commutators in A satisfy the Jacobi identity, so ρ kills every Jacobiator: $\rho(J(X^1, X^2, X^3)) = \pm 12\rho(W) = 0$, forcing $\rho(W) = 0$ and, by the locked normalization, $\rho(\{X^a, P_b\}) = 0$ too. Associativity cannot be imposed on this algebra at any nonzero $\hbar$ — the Jacobiator and the canonical pairing share the same W, so killing one classicalizes the whole structure, not just its non-associative part.

[DEF] For $s \in [0, 1]$, let M_s have brackets $\{X^a, X^b\} = 2s\,\epsilon^{abc} P_c$, $\{X^a, P_b\} = -2\delta^a_b W$, $\{P, P\} = 0$, W central — the Inönü–Wigner contraction of $M = M_1$ (Orbit B, §2) along weights $(1, 1, 2)$ on (X, P, W).

[FACT, proven; verified] The R-flux bracket and the Jacobiator vanish linearly in s while the Heisenberg bracket stays exactly constant (verified: $|\{X^1, X^2\}| = 2s$, $|J(X^1, X^2, X^3)| = 12s$ at $s = 1, 0.1, 0.01, 0$, the W-coefficient of $\{X^1, P_1\}$ bit-identical at -2.0000 throughout). M_0 satisfies the Jacobi identity on all 35 basis triples: it is the ordinary 3+3+1 Heisenberg Lie algebra, W alive and central. The obstruction above is evaded, not violated — the algebra itself has changed, so the no-go no longer applies, and M_0's associative representation theory (Stone–von Neumann) carries $W = \hbar \cdot 1 \neq 0$ directly. W is not transported as an independent generator; it is regenerated on arrival, $W = -\frac{1}{2}\{X^1, P_1\}$.

[FACT, proven] This is not a separate construction bolted onto Orbit B: M_0 is *exactly* Orbit A's algebra, obtained directly as a first-order $(1, 2, 3)$ contraction when the coordinate triple is the closed Fano line instead of the momentum triple. Orbit A and Orbit B are two stations of one tower:

$$\mathrm{Im}(\mathbb{O}) \xrightarrow{(1,2,3),\ \text{Orbit B}} M_1 \ (\text{R-flux Malcev}) \xrightarrow{(1,1,2)} M_0 \ (\text{Heisenberg}).$$

[OPEN, named explicitly] This document has not found, anywhere in the cited chain, a proof that the $(1, 1, 2)$ weight scheme is the *unique* second-contraction choice achieving this — only that it works, exactly and verifiably. This is structurally the same kind of question §2 answered for the *first* contraction (via the Weimar-Woods validity condition, exhaustively, over all 140 candidate gradings) — but no analogous exhaustive-validity argument has been located for the second contraction's weight choice specifically. That the endpoint coincides with the independently-established Orbit A is suggestive corroboration, not a substitute for such a proof. §13 returns to this, alongside two further items this compilation surfaces.

[OPEN, located in review correspondence, verified computationally] The two-station tower above invites a sharper question than "why pass through R-flux at all," and it has a precise, checkable answer at the structural level, though not yet at the physical one. Fixing the same seven octonion directions and the same Fano line throughout, Orbit A's route (X the closed line, direct $(1, 2, 3)$ contraction) and Orbit B's route (through R-flux, then the second contraction above) reach the identical Heisenberg algebra M_0 — the same pair of triples end up canonically paired by W either way, verified directly against the multiplication table. What differs between the two routes is *which* triple is identified as X: Orbit A's route makes X the triple that was a genuine closed Fano line — associative from the start; Orbit B's route makes X the triple that was *not* closed, the one that carried the non-associative content at the intermediate R-flux stage. This is a fact about the construction, confirmed directly, not a conjecture. What it leaves open is why physical space should be identified with the triple that was non-associative at the first stage rather than the one that was already associative — a question this document does not attempt to settle. One candidate answer was checked, and the check itself needed a second pass to get right: R-flux's own founding paper (Günaydin, Lüst & Malek) does *not* settle this either way at the title level — "missing momentum modes" names the fourth momentum absent from the toy M-theory background, the reason the phase space is seven-dimensional at all, not a claim about which surviving triple carries non-associativity, and citing the title against the spatial identification was a locator-level error, corrected here. What the founding literature's own body convention actually does, checked directly against the standard R-flux commutator $[x^i, x^j] \propto R^{ijk}p_k$ recurring across the string-theory literature: it assigns the surviving non-associative self-bracket to the *position* coordinates specifically, matching $\{X, X\} \propto P$ with X already named as coordinates — supporting Orbit B's identification at the level of convention. Convention is not derivation, so the physical question stays open — but for that reason, not because the founding literature cuts against it; if anything, the founding convention is a data point *for* the identification this document's own X-labeling already uses, not against it. That locator debt is now closed, checked directly against GLM's body equations rather than the abstract alone, and the result sharpens the evidence considerably. Two correspondences exist in GLM's paper, not one, and they carry different evidentiary weight. The directly relevant one — since it is built on exactly this document's own $1+3+3$ shape, not the four-coordinate M-theory extension — is GLM's own string-level contraction (their §3.2, presented as a direct demonstration, "we will now show that…", not hedged as a conjecture): $p_i \propto e_i$ for the closed quaternion triple $\{e_1, e_2, e_3\}$, and $x^i \propto f_i = e_{i+3}$ for the complementary triple $\{e_4, e_5, e_6\}$, with the central element $\propto e_7$. Checked directly against GLM's own multiplication table: $\{e_1, e_2, e_3\}$ is closed under the bracket (a genuine quaternion subalgebra, landing within itself), while $\{e_4, e_5, e_6\} = \{f_i\}$ is not (its bracket lands on $\{e_1, e_2, e_3\}$, outside the triple) — and, after contraction, it is exactly this open triple that carries the surviving nonzero associator, targeting the central element, matching this document's own $\{X, X, X\} \propto W$ structure exactly. GLM's own naming is therefore unambiguous and explicit, not a side effect of a commutator sign: they call the *non-closed*, associator-carrying triple the *coordinates*, and the closed, associative triple the *momenta* — exactly Orbit B's identification, stated as GLM's own deliberate choice rather than read off a convention. This is stronger evidence than the commutator check above, though still not a derivation from anything internal to this document's own construction: GLM's identification traces back to string-worldsheet results establishing the R-flux coordinate algebra independently, not to anything this document's own construction supplies. (The separate four-coordinate, three-momentum split this note previously pointed to belongs to GLM's *M-theory* extension specifically, which they explicitly flag as their own conjecture — a different, weaker-hedged correspondence than the string-level one above, and not the directly relevant one for this document's own $1+3+3$ shape.) The physical question — why this identification, rather than its reverse, should hold for the universe this construction describes — remains exactly as open as before; what has changed is the weight and precision of the evidence on one side of it.

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4. The dilation, created and split

[FACT, proven] The grading dilation $\sigma_\lambda : (X, P, W) \mapsto (\lambda X, \lambda^2 P, \lambda^3 W)$ is *not* a symmetry of the parent $\mathrm{Im}(\mathbb{O})$ for any $\lambda \neq \pm 1$: the closed triple's own nonzero bracket forces $\lambda^2 \cdot \lambda^2 = \lambda^2$, generically false. It *is* an exact automorphism of every M_s, $s \in [0, 1]$ — verified numerically, automorphism defect exactly 0 across tested (s, λ) pairs — since both surviving bracket families are exactly graded (1+1=2, 1+2=3). The dilation symmetry is *created* at the first contraction, precisely where the obstruction to it (the closed triple's own internal bracket) is killed.

[FACT, proven] On the fixed-W (Weyl) algebra — any representation with $W = \hbar \cdot 1$ — σ_λ fails to be an automorphism by exactly the factor λ^3 (verified: the bracket coefficient scales to $\lambda^3 \cdot (-2)$), and factors uniquely as

$$(\lambda, \lambda^2, \lambda^3) = \underbrace{(\lambda^{3/2}, \lambda^{3/2}, \lambda^3)}_{\text{rescales } W} \circ \underbrace{(\lambda^{-1/2}, \lambda^{1/2}, 1)}_{\text{squeeze: preserves } W},$$

both factors verified as exact automorphisms of their respective structures. The **squeeze flow**, generator $D = \frac{1}{2} \sum_a (X_a P_a + P_a X_a)$ in any representation, is the component of the octonionic dilation that fixed-$\hbar$ quantum mechanics can carry as an internal symmetry; the W-rescaling component is not an internal symmetry of any fixed-$\hbar$ theory.

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5. The $\mathfrak{so}(3)$ remnant of G_2

[FACT, proven] Fixing the grading breaks $G_2 = \mathrm{Aut}(\mathbb{O})$ to a computable remainder. Derivations of a composition algebra are skew with respect to the norm form (Schafer), so a grading-preserving derivation D satisfies $D(e_4) \in \mathbb{R}e_4$ and $\langle D(e_4), e_4 \rangle = 0$, forcing $D(e_4) = 0$; the Leibniz rule then gives $D(e_4 x) = e_4 D(x)$, so D commutes with the complex structure $J = L_{e_4}$ on the six-dimensional complement, and preserving the P-triple as a real subspace forces D's matrix real, hence in $\mathfrak{so}(3)$, acting identically on $X = J(P)$.

Two stages, predictions stated before computation: **Stage 1**, the stabilizer of e_4 alone — predicted $\mathfrak{su}(3)$ ($G_2/SU(3) = S^6$, classical) — computed dimension 8, matching. **Stage 2**, the full grading stabilizer — predicted $\mathfrak{so}(3)$ — computed dimension 3, matching, with the P-block antisymmetric, the X-block identical to the P-block, e_4 annihilated, and the three surviving generators closing under the bracket.

[FACT, proven; verified] This diagonal $\mathfrak{so}(3)$ — one rotation applied simultaneously to X and P, W fixed — is an automorphism of every M_s: the structure constants of M_s are built solely from the two $SO(3)$-invariant tensors δ_{ab} and ϵ_{abc}, so invariance is immediate from $R^T R = 1$, $\det R = 1$. Verified numerically for a random rotation: automorphism defect 1.3×10^{-15} at both $s = 1$ and $s = 0$. Within this construction — and given the ansatz of §2, three coordinate and three momentum directions to begin with, now a proven consequence of non-associativity retention rather than an assumption (§13, [OPEN 3], resolved) — the joint rotational isotropy *among* those three directions is *derived* as the unbroken octonionic remnant, not postulated. The symmetry-breaking chain: G_2 (14) $\supset \mathfrak{su}(3)$ (8) $\supset \mathfrak{so}(3)$ (3).

[FACT, proven] The squeeze flow and this $\mathfrak{so}(3)$ commute exactly. This follows directly, not merely numerically: §7–8 below establish that the squeeze acts as pure translation on the radial factor $L^2(\mathbb{R}_u)$ of the one-particle space $H_1 = L^2(\mathbb{R}_u) \otimes L^2(S^2)$, while rotations act purely on the angular factor $L^2(S^2)$ — operators of the form $A \otimes I$ and $I \otimes B$ on a tensor product always commute, by elementary tensor algebra, independent of what A and B individually are. (Numerically confirmed too, $\|[S_t, R \oplus R]\| = 0$ to machine precision — but the tensor-factor argument is the actual proof; the source paper's own gloss for this fact, "both are block-scalar," is corrected here rather than repeated: a genuinely block-scalar action would be abelian, and $\mathfrak{so}(3)$ is not. The commutation is real; that particular mechanism-gloss for it was not.)

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6. The state problem, and three theorems that shape it from outside

The two-contraction tower terminates in M_0, the $3{+}3{+}1$ Heisenberg algebra, with $W = \hbar \cdot 1$ fixed by §2's centrality theorem and Schur's lemma. Two structures survive: the squeeze flow (§4) and the $\mathfrak{so}(3)$ remnant (§5). **The state problem:** find ω KMS for the squeeze flow and invariant under the remnant.

[FACT, cited] Three theorems shape this from outside, each standard and each cited rather than re-derived here: no infinite-dimensional simple exceptional Jordan algebra exists (Zelmanov), keeping octonionic content finite and local; a JBW algebra is the self-adjoint part of a von Neumann algebra exactly when it admits a *dynamical correspondence* (Alfsen–Shultz) — dynamics must be state-borne, not algebra-resident; and a factor is semifinite exactly when its modular flow is inner (Takesaki), so at a type III target no algebra-resident density can generate the sought flow — the state must be characterized by the KMS condition directly, not read off a density matrix.

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7. The no-go, and the escape to the scale line

[FACT, proven] On any *finite* number of CCR fibers, no normal state is invariant under the squeeze flow, hence none is KMS for it. A normal invariant state is a positive trace-one operator commuting with e^{itD}; trace-one forces compactness, hence pure-point spectrum on finite-rank invariant subspaces on which D would need eigenvalues — but D's spectrum over n finite fibers is purely absolutely continuous (established below), so no such subspace exists, forcing the operator to zero, contradicting trace-one.

[FACT, proven] No invariant quasi-free covariance exists either, for any number of static fibers: solving exactly for the space of invariant symmetric forms (dimension 1 at $n = 1$, dimension 9 at $n = 3$, all cross-pairings included), every solution has identically vanishing x–x block, and the quasi-free positivity bound $\eta(x,x)\eta(p,p) \geq \frac{1}{4}\sigma(x,p)^2$ with $\sigma(x,p) = -2W$ then reads $0 \geq W^2$ — violated for every $W \neq 0$. The obstruction runs exactly through $W \neq 0$: the same structural $\hbar$ whose survival through two contractions was the whole point is what expels the sought thermal flow from every type-I setting, an internal counterpart of Takesaki's inner-flow-iff-semifinite criterion.

[FACT, proven; verified] The escape: the fiber squeeze is unitarily conjugate to rigid translation along the log-radial scale line. With $u = \log r$, $V : L^2(\mathbb{R}_+, dr) \to L^2(\mathbb{R}, du)$, $(Vf)(u) = e^{u/2} f(e^u)$ is unitary and intertwines the squeeze $(U_t f)(r) = e^{t/2} f(e^t r)$ with translation $u \mapsto u + t$; on the full fiber, $(Wf)(u,\theta) = e^{3u/2} f(e^u\theta)$ does the same with the three-dimensional Jacobian, carrying the squeeze to translation identically on every angular channel, generator $-i\partial_u$. Verified: the change-of-variables identity to defect 1.2×10^{-15}; unitarity by quadrature, $\|Vf\|^2 = 0.7106535$ against $\|f\|^2 = 0.7106546$ (difference: window truncation). **The spectrum of D is forced, not chosen: purely absolutely continuous, equal to all of $\mathbb{R}$.** No flow that merely *squeezes* static modes can be invariantly stated; a flow that *translates* along a continuum can — and the fiber supplies exactly that continuum.

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8. The construction

[DEF, with full functional-analytic detail] Per angular channel, the test space is $\mathcal{D}_0 = \{f : \mathbb{R}_u \to \mathbb{R} \text{ Schwartz},\ \hat{f}(0) = 0\} = \partial_u \mathcal{S}(\mathbb{R}, \mathbb{R})$ — a *membership* condition (the natural current test space), not a restriction on some larger physical space; $\mathcal{D}$ is the algebraic direct sum over angular channels. The symplectic form is $\sigma(f,g) = 2\int_0^\infty \operatorname{Im}(\hat{f}(p)^*\hat{g}(p))\,dp$, summed over channels. $\mathcal{D}$ is invariant under translation and the $SO(3)$ channel action, both preserving σ (proven).

The one-particle structure: $K : \mathcal{D} \to H_1 = L^2((0,\infty), dp) \otimes (\text{angular})$, $Kf = \hat{f}|_{p>0}$, satisfying $2\operatorname{Im}\langle Kf, Kg\rangle = \sigma(f,g)$ with $K\mathcal{D} + iK\mathcal{D}$ dense in H_1 — a genuine Kay–Wald one-particle structure. The one-particle generator h = multiplication by p is self-adjoint, strictly positive, absolutely continuous, trivial kernel — forced by the Mellin conjugacy of §7, not chosen.

The observable algebra is the Weyl C*-algebra $\mathcal{W}(\mathcal{D}, \sigma)$ (Manuceau–Sirugue–Testard–Verbeure) — the smallest C*-algebra for these canonical commutation relations; dynamics and rotations are the induced Bogoliubov automorphisms.

[DEF] ω_β is the quasi-free state with two-point function $\mathcal{K}(f,g) = \int_0^\infty [(1+\rho(p))\hat{f}(p)^*\hat{g}(p) + \rho(p)\hat{f}(p)\hat{g}(p)^*]\,dp$, $\rho = (e^{\beta p} - 1)^{-1}$, summed over channels.

[FACT, proven] Every property needed is in hand. *Convergence*: infrared, $f \in \mathcal{D}_0 \Rightarrow \hat{f}(p) = O(p)$, so the thermal density is $O(p/\beta)$, integrable at 0 — the membership condition doing exactly this job; ultraviolet, Schwartz decay. *Positivity*: the kernel is a sum of two positive-semidefinite Gram forms (weights $1+\rho > 0$ and $\rho > 0$); verified concretely on 14 random current-smeared functions, hermiticity defect 1.4×10^{-14}, minimum eigenvalue $+1.36 \times 10^{-2}$. *Regularity*: $\mathcal{K}(f,g)$ continuous in its arguments, so the Weyl groups are weakly continuous and the fields genuinely exist. *KMS*: with $F(z) = \omega_\beta(\phi(f)\alpha_z\phi(g))$ — α the dynamics, kept distinct from the split involution τ of §1 — analyticity on the strip $0 \le \operatorname{Im} z \le \beta$ follows from the damping/growth bounds on each term, and the boundary identity $F(t+i\beta) = G(t)$ follows in three lines from $(1+\rho)e^{-\beta p} = \rho$ and $\rho e^{\beta p} = 1+\rho$ — verified numerically to 1.1×10^{-16} at several t. *Invariance*: dynamics- and $SO(3)$-invariance both immediate, ρ being a function of h alone.

ω_β is KMS for the squeeze flow by §7's conjugacy (the same operator, different coordinates); $SO(3)$-invariant automatically (§5's commutation); $W = \hbar \cdot 1$ uniformly. Nothing is inserted by hand: the algebra came from the tower, the flow from the dilation's $\hbar$-preserving shadow, the mode continuum from the Mellin conjugacy, the state from the KMS condition itself.

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9. The Kähler structure underlying the covariance

Provenance, stated once and plainly: this section's specific construction is not drawn from the archive, though one piece of its raw material is. The complex structure $J = L_{e_4}$ is not new here — it is already named and used in the second-contraction paper, for the $\mathfrak{so}(3)$-remnant derivation §5 already cites. What §9 builds from it is new: the metric $g := \sigma(\cdot, J\cdot)$, its survival through both contractions, and the decomposition of $\mathcal{K}$. The section's underlying doctrine — symmetric half as metric, antisymmetric half as $\hbar$ — is also prior art, checked directly: *The Two Halves of One Product: The Jordan/Malcev Split of the Octonions and the Measurement Channel* proves exactly this split at the algebra level (its own Proposition 1: the Jordan half of an imaginary octonion product is purely scalar, $-\langle a, b\rangle$; the Malcev half is purely imaginary) and names it "two scalar emissions." That paper stops at the conceptual split — no Kähler triple, no complex structure invoked for this purpose, no covariance decomposition — checked directly rather than assumed. The immediate trigger was an informal observation raised by Harald Kjøge Rønning in correspondence external to the numbered papers — that octonion multiplication of two imaginary elements "spits out" a real scalar, distinct from any of the emergent spatial directions — which the archive had already proven in general form without this document connecting it to J, g, or $\mathcal{K}$. The Kähler-triple formalization built from these pieces, and the verification below, were carried out within this project's own review process. It is the chain's first original extension built from prior material, not a restatement of any single prior document, and is held to the same evidentiary standard as everything drawn from the archive: proved, verified, or flagged open, with nothing asserted on the strength of where the idea came from.

The construction above builds the operational algebra and its thermal state without reference to any metric structure on $\mathrm{Im}(\mathbb{O})$ beyond the bracket itself. This section closes a gap the construction leaves open: whether the covariance $\mathcal{K}$ of §8 — and specifically its dependence on β — traces back to the octonionic seed with the same precision as everything else in this chain, or whether it is simply postulated once the algebra is in hand. It does trace back, precisely, in three independently checked steps.

[DEF] On $\mathrm{Im}(\mathbb{O})$, define a symplectic form $\sigma(x, y)$ as the mediator-direction (e_4) component of the commutator $[x, y]$; a complex structure $J := L_{e_4}$, left multiplication by the mediator; and a candidate metric $g(x, y) := \sigma(x, Jy)$. This is the standard compatible-triple construction of symplectic geometry, and it recovers the octonionic real part exactly: for imaginary x, y, $xy = -\langle x, y\rangle + x \times y$, so $\mathrm{Re}(xy) = -\langle x, y\rangle$, and g built this way reproduces $-\mathrm{Re}(xy)$ up to the same overall normalization σ itself carries.

[FACT, proven; verified] Restricted to $X = \{e_5, e_6, e_7\}$ with $J(X^a) := e_4 \cdot X^a$ ($J(X^1) = e_4 e_5 = e_1$, etc.), $g(X^a, X^b) = \sigma(X^a, J(X^b))$ — the e_4-coefficient of $[X^a, J(X^b)]$ — equals $2\delta^{ab}$ exactly, positive-definite as a compatible triple requires: verified against the full multiplication table, all three diagonal pairs, with off-diagonal pairs vanishing exactly rather than approximately. This computation is self-contained and does not need to match §2's separately-introduced P_a sign-for-sign; checked fully against all three of §2's own bracket families simultaneously ($\{X, X\}$, $\{X, P\}$, and the validity of $\{P, P\} = 0$ after contraction, not before), §2's own labeling under this same straight indexing is $P_a = -e_a$ and $W = +e_4$ — so $J(X^a) = +e_a = -P_a$: this section's $J(X^a)$ and §2's P_a are each internally consistent within their own definitions and differ from each other by exactly this sign, which is why they need not, and do not, have to coincide for either to be correct.

[FACT, proven] g survives both contractions of §§2–3 unchanged. The mechanism is grade-matching, not coincidence: for a bracket $\{X, P\}$ pulled back through a contraction with weights $(a, b, a+b)$ on (X, P, W), the input grade $a + b$ exactly matches W's own assigned weight for *both* contractions used in this chain — $(1, 2, 3)$ for the first, $(1, 1, 2)$ for the second — so the rescaling factors in the pullback cancel exactly at every stage, for both. (A naive check that instead evaluates the raw bracket on rescaled arguments and reads off the mediator component will appear to show degeneration at any weight scheme whatsoever, first and second contraction alike; this is an artifact of bilinearity, not a statement about contraction survival, and is not the correct object to inspect. The correct check tracks the coefficient of the *contracted* algebra's own central generator, which is what "the pairing survives independently of the contraction parameter" already means by construction.) Consequently g, computed on the finite Heisenberg generators X^a of §3, is identical to g computed on the raw octonionic X^a of $\mathrm{Im}(\mathbb{O})$ — the same $2\delta^{ab}$, unconditionally.

[FACT, proven; verified] For any fixed $p > 0$, $\hat{f}(p)$ is a single complex number, and $2\,\mathrm{Im}(\hat{f}(p)^*\hat{g}(p))$ is exactly the standard symplectic form on that number's real and imaginary parts — verified symbolically. Keystone's $\sigma(f, g) = 2\int_0^\infty \mathrm{Im}(\hat{f}(p)^*\hat{g}(p))\,dp$ of §8 is accordingly not an analogy to a continuum of finite canonical pairs; it is built as one, literally, integrated over p.

[FACT, proven; verified] Decomposing $\mathcal{K}(f, g)$ of §8 into real and imaginary parts (using $\hat{f}(p)\hat{g}(p)^* = \overline{\hat{f}(p)^*\hat{g}(p)}$) gives, exactly:

$$\mathcal{K}(f, g) = \int_0^\infty \coth(\beta p/2)\,\mathrm{Re}(\hat{f}(p)^*\hat{g}(p))\,dp \;+\; i\int_0^\infty \mathrm{Im}(\hat{f}(p)^*\hat{g}(p))\,dp,$$

verified symbolically and numerically. The imaginary part is exactly $\sigma(f, g)/2$, entirely independent of β — matching, at the field-theoretic level, the same β-independence of σ already established for the finite generators above. The real part carries the entire thermal content, as the single multiplicative weight $\coth(\beta p/2) = 2\nu(\beta, p)$ applied to the vacuum ($\beta \to \infty$) inner product $\int \mathrm{Re}(\hat{f}^*\hat{g})\,dp$. β does not modify the underlying (σ, J, g) geometry; it selects, among the family of states compatible with that fixed geometry, the one satisfying the KMS condition at temperature $1/\beta$, and this selection acts as a pure reweighting of the metric part alone.

[FACT, proven; verified] The finite $\mathfrak{so}(3)$-vector structure of g identifies with the $l{=}1$ angular sector of the "angular" factor in $H_1 = L^2((0, \infty), dp) \otimes (\text{angular})$, not merely by analogy but as the identical representation. Writing rotation of a function on S^2 as $(R \cdot f)(\theta) = f(R^{-1}\theta)$ and restricting to the three linear functions $\theta^1, \theta^2, \theta^3$, all three standard $\mathfrak{so}(3)$ generators act on this three-dimensional space of functions by exactly the same matrices as they act on the vector $(\theta^1, \theta^2, \theta^3)$ directly — verified for all three generators, zero difference. Closure is exact rather than approximate: since θ^a is linear in θ and R^{-1} is a linear map, $(R^{-1}\theta)^a$ remains exactly linear under any rotation, with no admixture from other angular-momentum sectors, by construction rather than by limit. That $\theta = X/|X|$ is the direction of the same X whose $\mathfrak{so}(3)$ action §5 already derives means this identification is forced by what θ *is*, not a separate structure requiring its own justification.

[OPEN, the channel-normalization convention] One piece of bookkeeping is not settled by anything above. The $L^2(S^2)$ inner product of the raw functions θ^a is $\langle \theta^a, \theta^b \rangle = \frac{4\pi}{3}\delta^{ab}$ — verified by exact symbolic integration — while the orthonormalized real spherical harmonics satisfy $\langle Y_1^a, Y_1^b \rangle = \delta^{ab}$

by definition. Both sides of the identification above agree exactly on *shape*: g and this Gram matrix are each exactly proportional to δ^{ab}, with no preferred direction on either side — a convention-independent fact, since isotropy is a basis-independent property. They need not agree, and this document does not fix which they should, on *overall scale*: whether "angular channel a" in §8's construction means the raw θ^a or the orthonormalized Y_1^a is not stated explicitly there. The way §8's own σ formula is written — summed over channels with no extra numerical factor — is suggestive of the orthonormalized convention (raw, unnormalized channels would need to carry the $\frac{4\pi}{3}$ explicitly, and it is not present), and g's own coefficient is purely algebraic, with no π anywhere in its derivation, which would sit oddly against an implicit geometric factor if the raw convention were intended instead. This is an argument for which convention is meant, not a proof that no other is possible; the honest fix is one explicit sentence in §8 stating the convention, not yet written there.

Summary, against the three-part scoping this section answers. Whether g survives the algebra's own contractions: it does, exactly, for a stated structural reason. Where β enters the covariance: at one identified point, as a multiplicative weight on the metric part alone, leaving the symplectic structure untouched. Whether the finite vector structure and the field-theoretic angular structure are the same representation: they are, verified generator by generator. One item remains open, named above, and it is a convention to be stated rather than a further computation to be run.

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10. The type computation

[FACT, cited] (Standard, Connes.) $S(M) \setminus \{0\}$ is a closed multiplicative subgroup of $\mathbb{R}_+^\times$; the trichotomy $\{1\}$ / $\lambda^{\mathbb{Z}}$ / $\mathbb{R}_+^\times$ is the type trichotomy semifinite / III_λ / III_1.

[FACT, cited] (Standard, Araki.) For the quasi-free KMS state, $\log \text{Spec}(\Delta_\omega)$ is the closed additive group generated by $\beta \cdot \text{supp}(h)$ under finite particle/hole insertions.

[FACT, verified] The generated group was constructed directly for four spectral hypotheses (grid resolution 10^{-5}, $\beta = 1$): a single energy freezes on a lattice (III_λ); the octonionic grading ladder $(1, 2, 3) \cdot 0.7$ likewise freezes, $\lambda = e^{-0.7} \approx 0.4966$; two incommensurate energies densify toward III_1; and the forced interval $(0, 1]$ — this construction's actual case — saturates $\mathbb{R}$ outright, gaps shrinking to 10^{-5} by 32 insertions. Corollary 1.1 (§7) forces absolutely continuous spectrum on all of $\mathbb{R}$, so the generated closed group is $\mathbb{R}$ and $\text{Spec}(\Delta_\omega) = [0, \infty)$.

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11. The centralizer, closing the last construction-specific link

[FACT] Passing from $\text{Spec}(\Delta_\omega)$ to Connes' invariant $S(M) = \bigcap_\varphi \text{Spec}(\Delta_\varphi)$ needs Connes' Théorème 3.2.1: $S(M) \cap \mathbb{R}_+^\times = \Gamma(\sigma^\varphi) = \bigcap\{\text{Spec}(\Delta_{\varphi_e}) : e \in \text{Proj}(M_\varphi), e \neq 0\}$, an intersection over the *centralizer* M_φ. A trivial centralizer collapses this to the single term already computed.

[FACT, proven] $M_{\omega_\beta} = \mathbb{C}1$. For a β-KMS state, the modular flow is the dynamics reparametrized, $\sigma_t^\omega = \alpha_{-\beta t}$ (standard; α the dynamics, distinct from the split involution τ of §1), so M_ω is exactly the fixed-point algebra of α. Let $x \in M_{\omega_\beta}$; then $x\Omega$ is invariant under the implementing unitary group $U(t)$. In the Araki–Woods GNS form of a quasi-free KMS state, $U(t) = \Gamma(e^{it\ell})$ with one-particle Liouvillean $\ell = h \oplus (-h)$ over the doubled space, particles and holes. Since $\text{Spec}(h) = [0, \infty)$ is purely absolutely continuous with no zero eigenvalue (§7's Mellin conjugacy), ℓ has no eigenvalues, and on every n-particle sector ($n \geq 1$) the spectral measure of $d\Gamma(\ell)$ is an n-fold convolution of absolutely continuous measures, hence itself absolutely continuous — no eigenvector, so no invariant vector, in any such sector. Only the vacuum sector remains: $x\Omega = c\Omega$, and since ω_β is faithful, Ω is separating, forcing $x = c1$. □

[FACT, proven] Three consequences follow immediately. *Factoriality*: modular automorphisms fix the center pointwise, so $Z(M) \subseteq M_{\omega_\beta} = \mathbb{C}1$ — M is a factor, derived rather than assumed. *The link closed*: $S(M) \cap \mathbb{R}_+^\times = \mathrm{Spec}(\Delta_{\omega_\beta}) \cap \mathbb{R}_+^\times = \mathbb{R}_+^\times$, giving type III_1 outright, resting on no further construction-specific citation. *Canonicity*: granting injectivity of the quasi-free algebra (closed by a recorded standard argument in §13, not asserted), Haagerup's uniqueness theorem — stated for exactly this hypothesis — identifies M as *the* injective type III_1 factor.

[FACT, verified] The $\mathrm{III}_1/\mathrm{III}_\lambda$ fork is visible a third, independent way, in the centralizer itself: in the lattice counterfactual (energies on the octonionic grading ladder), the two-particle Liouvillean has an atom at zero of exact mass $\frac{1}{3}$, giving genuine invariant multi-particle vectors and a nontrivial centralizer; in the forced continuum case, the mass of $\{|p_1 - p_2| < \epsilon\}$ vanishes linearly (computed: mass$/\epsilon \to$ 2.000 as $\epsilon \downarrow 0$) — no atom, no invariant vectors, trivial centralizer, confirming the theorem above by an independent, dynamical route.

[FACT, verified] Supporting checks close remaining gaps at the one-particle level: the thermal covariance $S(p) = \frac{1}{2}\coth(\beta p/2)$ satisfies $S(p) - \frac{1}{2} \geq 0$ exactly, upgrading existence to the elementary quasi-free construction; thermal correlations under current smearing decay cleanly ($0.540 \to$ 1.6×10^{-4} over $t = 0$ to 80); and the infrared condition is *quantitatively necessary* — without it the two-point integral diverges logarithmically, increments matching the predicted $\beta^{-1} \ln 100 = 4.6052$ to four decimals.

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12. Uniqueness

[FACT, proven] ω_β is the unique regular β-KMS state of the squeeze flow on this algebra, *outright* — rotation-invariance is not an independent constraint but a derived consequence. Within the full quasi-free class (anomalous pairs and channel mixing permitted), the KMS boundary condition on the two-point function yields two operator equations. The anomalous equation, $(1 - e^{-\beta(p+q)})A(p,q) = 0$, has a strictly positive coefficient for all $p, q > 0$ (verified, minimum 2.0×10^{-4} at the grid edge), forcing $A \equiv 0$ — no squeezed KMS states exist. The invariant equation, $\rho(e^{\beta h} - 1) = 1$, has a unique solution since $h > 0$ with trivial kernel makes $e^{\beta h} - 1$ invertible: $\rho = (e^{\beta h} - 1)^{-1}$, exactly the Planck operator already used (recovered to 4.3×10^{-9}). The unique solution is a function of h alone, so channel-mixing is excluded automatically, and since h commutes with $SO(3)$, rotation-invariance is a *consequence* of thermality, never an independent input.

[FACT, proven] Beyond the quasi-free class, the classical KMS classification for free Bose dynamics (Rocca–Sirugue–Testard) locates every non-uniqueness mechanism at the zero mode — condensate or chemical-potential families. This door is shut twice: spectrally, a condensate displacement needs an element of $\ker h$, and $\ker h = \{0\}$ by the Mellin conjugacy; algebraically, the infrared-smeared domain removes the zero mode from the algebra entirely, so the would-be shift automorphisms act trivially. The computation makes this one fact, not two: the uniqueness equation's coefficient $e^{\beta p} - 1$

vanishes only at $p = 0$ — exactly the point the infrared condition excises.

[FACT] The result interlocks with §11: trivial centralizer makes ω_β a factor state, hence an extreme point of the KMS simplex — and the uniqueness argument above shows there are no other extreme points either. The simplex is the single point $\{\omega_\beta\}$.

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13. What remains genuinely open

Three items, named precisely rather than absorbed into the chain's confidence — one substantially closed on inspection, one located by the act of compiling this chain in one place and since resolved by a companion paper, one a declared conditionality worth more weight than a ledger line.

[OPEN 1, second contraction's weight scheme — relocated, not closed] §3 proves the $(1, 1, 2)$ contraction achieves the transition from R-flux to Heisenberg, exactly and verifiably, and that its endpoint coincides with the independently-classified Orbit A. As an *endpoint-reaching* question this is badly underdetermined: normalizing the X-weight to 1, every scheme $(1, b, 1+b)$ with $b < 2$ reaches Heisenberg exactly (the canonical pairing survives s-independently, since $a + b - (a + b) = 0$ regardless of b; the R-flux bracket dies, since its decay exponent $2a - b = 2 - b > 0$ throughout the range) — a continuum, not a discrete choice, so "which weights reach the target algebra" is close to vacuous as a question.

But the endpoint is not all that matters, by this document's own §4: a contraction *creates a dilation*, and the created dilation *is* the weight scheme itself, (s^a, s^b, s^{a+b}), acting on whatever the contraction lands on. Running §4's own $\hbar$-preserving factorization on this dilation — exactly the computation already used there for the *first* contraction, not a new technique — gives a W-rescaling component and a squeeze component $(s^{(a-b)/2}, s^{(b-a)/2}, 1)$, trivial if and only if $a = b$ (verified symbolically; the $a = 1, b = 2$ case reproduces the already-established first-contraction exponents $(-1/2, 1/2)$ exactly, confirming the same formula is being applied consistently, not a new one introduced for this case). Every scheme with $a \neq b$ therefore creates a *second, independent squeeze flow* at the second contraction, on top of whatever the first contraction already created; the scheme with $a = b$ — normalized, exactly $(1, 1, 2)$ — is the unique one that is dynamically *silent*: pure W-rescaling, no new flow content, so that the framework's single derived squeeze flow can be said to descend from the first contraction alone, with nothing added at the second.

[PROP, principle-relocated criteria, verified symbolically, not independently confirmed as physically necessary] Three superficially independent criteria converge on this same point. *Dynamical silence*, as above: $a = b$. *X/P symmetry*: the contraction that erases the first contraction's own X/P asymmetry is itself symmetric between X and P — again exactly $a = b$. *Minimal-integer degeneration*: among integer-weight schemes, the R-flux bracket's decay exponent $2a - b$ is minimized, at value 1 (the slowest allowed decay, rather than $2, 3, \ldots$), exactly at $b = 1 = a$. All three, conceptually distinct, select the identical scheme.

What this does and does not do. It does not discharge [OPEN 1]: "single source of dynamics" — the requirement that the second contraction introduce no flow content the first did not already supply — is a *principle*, stated and motivated here, not a theorem derived from anything more basic. The honest status was, until a companion paper resolved [OPEN 3], exactly parallel to [OPEN 3]'s own relocation: the free input moves from an unexplained numerical choice, $(1, 1, 2)$ among a continuum, to a named principle whose own justification is not yet supplied. [OPEN 3] has since been resolved (below): its residual input is no longer "why canonical pairing at all," which is now a proven consequence, but the sharper "why demand that non-associative content survive at all" — itself identified as this project's founding motivation rather than a further mystery. Whether [OPEN 1]'s principle and this residual input are secretly one requirement — both being versions of "the structure defining the theory enters exactly once" — remains a further, unresolved question this document does not attempt to settle.

[PROP, standard argument, closing OPEN 2] Injectivity of the quasi-free algebra constructed in §8 — the property Haagerup's own theorem is actually stated for (its title names *the injective factor of type III*$_1$, not "the hyperfinite one") — is closed here by a standard three-step argument, offered at the same evidentiary weight as the rest of this document: not re-invented, but assembled and checked, deliberately stopping at injectivity rather than reaching for the stronger-sounding but unnecessary "hyperfinite." Exhaust the test space $\mathcal{D}$ by an increasing net of finite-dimensional, symplectically nondegenerate subspaces $\mathcal{D}_n \subset \mathcal{D}$. For each $\mathcal{D}_n$, the von Neumann algebra $\pi_\omega(\mathcal{W}(\mathcal{D}_n, \sigma))''$ generated *in situ* — inside the one ambient GNS representation of ω_β on the full algebra, not a separately-built representation of the restriction — is **type I**: for finitely many modes, Stone–von Neumann forces every regular representation to be a multiple of the unique Schrödinger representation, and ω_β restricted to $\mathcal{D}_n$ remains regular (inherited from the same continuous Weyl family established for the full state in §8), so $\pi_\omega(\mathcal{W}(\mathcal{D}_n, \sigma))''$ is manifestly a multiple of $B(L^2(\mathbb{R}^n))$ — type I, hence injective, elementarily (type I algebras are direct sums/integrals of $B(H)$'s, and injectivity is closed under this). The union $\bigcup_n \mathcal{W}(\mathcal{D}_n, \sigma)$ is weakly dense in the full algebra, since every generating Weyl unitary $W(f)$, $f \in \mathcal{D}$, lies in some $\mathcal{D}_n$ as the net exhausts $\mathcal{D}$. **The step that closes it:** injectivity passes to the weak closure of an increasing net of injective subalgebras — standard and citable, via a weak-$*$ limit of the norm-one conditional-expectation projections onto each piece, not requiring anything beyond this. The full algebra is therefore injective. This is *not* the same claim as hyperfiniteness (AFD: an increasing union of genuinely *finite-dimensional* pieces, which these type I pieces — multiples of $B(L^2(\mathbb{R}^n))$ — are not); that stronger equivalence is Connes' 1976 classification theorem (*Ann. of Math.* 104), a deep result this argument does not need and does not invoke. Injectivity alone is the correct and sufficient target, since it is exactly Haagerup's own hypothesis. **If this argument is accepted, Haagerup's canonicity in §11 becomes unconditional rather than resting on an unclosed assertion.** It is recorded here at the honest weight of a standard argument assembled for this specific construction, not a citation to a source that states it for this case outright — a distinction the next reader should be free to weigh independently.

[OPEN 3, the grading ansatz — located by this compilation, resolved by a companion paper] §2's exhaustive classification (56 of 140 valid, 28 of those genuine) is airtight *given* the $1{+}3{+}3$ grading shape: one central generator, three coordinates, three momenta. That shape itself — including, specifically, the *three*-dimensionality of both the coordinate and momentum triples — is never derived anywhere in *this* chain; it is the shape used by the founding literature (Günaydin–Lüst–Malek) and inherited here without an argument for why no other shape (a $2{+}2{+}3$ partition, a shape with more than three grade levels) was tried and excluded, rather than simply not tried. One source paper's own scope note confirms this directly: Malcev-preservation's genericity is stated as established "for the $1{+}3{+}3$ grading shape specifically," with other shapes named as untested — untested *as candidate foundations for this construction*, that is; a further paper in the chain, the same one supplying §2's matching lemma, tested one such shape *structurally* instead, a different and weaker kind of test than building a full alternative construction on it. At four grade levels, where both self-brackets have somewhere to land, the clean coordinate-versus-momentum asymmetry of the $1{+}3{+}3$ case does not merely relax into optional flexibility — it inverts into a stricter, joint requirement: validity itself forces both self-brackets to survive together or neither to, confirmed with zero exceptions across the full search — stronger evidence still that the $1{+}3{+}3$ pattern is that specific shape's own geometry speaking, not a structural law the algebra imposes independent of shape, though it stops short of testing whether a full construction *could* be built on any alternative shape. This is the sharpest form yet of a question raised elsewhere in this project under "absences" — why three spatial dimensions, rather than some other number — and compiling the chain in one place is what made it visible as *exactly this*, rather than a diffuse worry.

[THM, cited — OPEN 3 resolved] The free input identified above is not, after all, free. A companion paper (*The Non-Associativity Selector*, cited below) proves, over the full space of gradings of $\mathrm{Im}(\mathbb{O})$ — any partition, any number of grade levels, any positive real weights, with no dependence on the enumeration that first found the result — that exactly two outcomes exist: the contraction is 2-step nilpotent (hence Lie, hence physically trivial as an R-flux candidate), or it realizes exactly the $1{+}3{+}3$ R-flux pattern used throughout this document. No third outcome exists, by an eight-step argument in Fano-plane incidence and grade arithmetic alone, independent of the census that located it. The threeness of space, downstream of everything in §§4–11, therefore traces back not to an unexamined default but to a single, named hypothesis — that the contraction retain any non-associative content whatsoever — which is not itself a further free choice: it is this project's founding motivation, the entire reason the seed is $\mathbb{O}$ rather than an associative algebra. Demand that content survive, and the central generator, the canonical pairing, and the three-dimensionality of each block arrive as consequences, not further inputs. Foundation's own citation of this construction should be re-scoped accordingly, in whatever revision next addresses it. (A statement about single-grading contractions; the second contraction's own scheme remains [OPEN 1]'s subject, above.)

[TARGET, OPEN 3's decidable first stage — resolved] The question was not open-ended, and it has now been decided. Admissible grading shapes on the seven-dimensional $\mathrm{Im}(\mathbb{O})$ form a small, enumerable family, each testable by the same Weimar-Woods validity machinery already implemented in §2, against the four properties this chain actually consumes: validity itself; surviving non-associative content; a central $\hbar$ candidate; and a viable downstream operational algebra. The companion enumeration paper ran this computation twice, independently, then went further and proved the result unconditionally: not merely that $1{+}3{+}3$ is the unique viable shape among those enumerated, but that no enumeration was needed to know it — every grading, over any partition, any number of levels, any positive real weights, contracts to either a Lie algebra or exactly this pattern. [OPEN 3]'s first stage discharges outright; the ansatz converts into a theorem.

[PROP, verified observation refining OPEN 3, resolved — the underlying intuition is Harald Rønning's own, an archive search having found no prior capture of it despite two independent queries] A structural refinement of the question, independent of the enumeration above and checkable in its own right: the bracket-closed subspaces of $\mathrm{Im}(\mathbb{O})$ have dimension exactly $\{0, 1, 3, 7\}$, and no other. Dimension 0 and 1 trivially; no closed 2-planes exist, since for orthonormal imaginary x, y, $[x, y] = 2xy$ is orthogonal to both (a standard octonionic fact) and nonzero (the octonions have no zero divisors), hence outside $\mathrm{span}(x, y)$ — verified here directly against an explicit multiplication table, all 21 basis pairs and a further 8 random orthonormal pairs, residual zero to machine precision in every case; every closed 3-space is $\mathrm{Im}(\mathbb{H}')$ for some quaternion subalgebra — pick orthonormal x, y within it, closure then forces xy in as well, and $\mathrm{span}(x, y, xy)$ is exactly a Fano line in the aligned basis — verified the same way, generic pairs included; and dimensions 4–6 are impossible, since a closed V containing a quaternion triple $\{x, y, xy\}$ plus any further orthogonal unit z is forced, by closure alone, to also contain xz, yz, $(xy)z$, and these seven vectors together span all of $\mathrm{Im}(\mathbb{O})$ — verified for all four remaining directions z against a fixed quaternion triple, rank 7 without exception, with the supporting alternativity identity $x(xz) = -z$ holding exactly. **What this changed, and what it sharpened into:** imposing only the canonical-pair shape itself — one central direction, coordinate and momentum blocks of *equal* dimension n (forced by nondegenerate symplectic pairing, a standard fact about symplectic vector spaces), exhausting all seven imaginary dimensions — gives $7 = 2n + 1$, hence $n = 3$ uniquely; and at $n = 3$, a closed momentum block must then be a Fano line by the dimension result above, from which §2's own machinery (external mediator, matching lemma, the 28-orbit) follows. This did not, on its own, close [OPEN 3]: it relocated the free input from "why three coordinate and three momentum directions" to "why a canonical pairing — conjugate variables — at all," one level further back, and sharpened the [TARGET] enumeration above into a two-part conjecture: *every viable grading of* $\mathrm{Im}(\mathbb{O})$ *is Heisenberg-shaped, and every Heisenberg-shaped grading is exactly the* $1{+}3{+}3$. Both halves have since been confirmed by the companion enumeration paper, and in a stronger form than conjectured: the second half holds with room to spare, since the paper's own scan of its full structural classification finds exactly two Heisenberg-shaped patterns among the entire landscape — Heisenberg itself and R-flux — with no third candidate anywhere to rule out; and the first half is subsumed rather than separately needed, since the paper's Theorem A/B dichotomy decides viability directly from non-associativity retention alone, without reference to the Heisenberg-shape criterion this conjecture's own first half was stated in terms of. "Three-dimensionality is equivalent to canonical pairing, given the seed" is accordingly not quite the final form of the result — the proven form is one level more economical still, and one-directional, not a biconditional: three-dimensionality *follows from* retaining any non-associative content at all, given the seed, and canonical pairing is what comes along for free once that single condition is imposed. Only in that direction: the corpus's own Orbit A is itself a three-plus-three-plus-one grading — the identical block shape, X and P each dimension three, one central W — whose contraction retains nothing (§2, §3: M_0 is exactly Orbit A's algebra, ordinary and fully associative Heisenberg). Three-dimensional block structure does not imply retention; retention implies everything, which is exactly why retention, not shape, is the fundamental hypothesis.

[FACT, honestly carried, and worth more than a ledger line] A further item is not a gap but a declared conditionality, and it deserves to be named as what it actually is: the first place in this entire corpus where a genuinely distinguishing, physically falsifiable prediction-shape appears (registry item REG-8). The III_1 verdict is conditional on thermal time running along the squeeze flow specifically. If instead the physically realized dynamics couples to the discrete octonionic grading ladder rather than the continuum this construction derives, the modular spectrum instead freezes to a lattice, giving type III_λ with $\lambda = e^{-\beta E_0}$ — a preferred scale, the contraction's own signature, imprinted directly and in principle readable off the modular spectrum. Both outcomes remember their origin; only one is what this document, and everything built on it, assumes — and unlike most of what this project carries as open, this fork is the kind experiments are eventually made of, however far downstream that lies.

[SCOPE] Every contraction in this chain, including the enumeration proposed above, is a construction device — a tool for building the operational algebra from $\mathrm{Im}(\mathbb{O})$ — not an asserted physical process the seed is claimed to undergo. Whether settling itself is contraction-like in nature is a separate, open mechanism question, entirely outside this chain's current scope: recording the question here is not the same as taking a position on it.

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14. What this document is

A sequential compilation, not a new result: every proposition above is drawn from a specific source paper, cited by content rather than by a bare title, checkable against the constructions and computations as reproduced. Nothing here is claimed to strengthen what the sources already establish; the purpose is exclusively to remove the need to hold ten separate documents in mind at once to see the whole chain from $(\mathbb{S}, \tau)$ to the injective type III_1 verdict, and to name, in one place, exactly where that chain was not yet closed. **The ten source papers, in the order this document uses them:** the split's classification (§1); W and the R-flux Malcev algebra (§2); the genuine-versus-trivial contraction classification, in two papers (§2); a fifth, initially thought unneeded and restored on inspection — the four-level-shape paper, whose §2 proves the Fano matching lemma §2's own incidence argument cites directly, and whose main result (the X/P asymmetry dissolves once a shape provides both self-brackets a target) is independent corroboration for [OPEN 3] below, not merely a source of one borrowed lemma; the second contraction, the dilation, and the $\mathfrak{so}(3)$ remnant (§§3–5); the squeeze-thermal-state construction and type verdict (§§6–8, 10); the centralizer paper (§11); the uniqueness paper, *The State Is Unique* (§12); and the infrared write-out paper, *The Infrared Write-Out* (§8). §9 is not drawn from any single one of these; it connects results already established across several of them (the octonionic Kähler structure of §5, the covariance of §8) that no source paper states as a single claim. An eleventh paper, written after this compilation and citing it directly for the question's precise statement, resolves [OPEN 3] (§13): *The Non-Associativity Selector*, whose theorem this document now cites rather than reproduces, on the same footing as §9 — the chain's second original extension, not a compilation of prior work, arrived at only because compiling the chain in one place is what first made the question visible as decidable. Foundation's own inventory names five of the original ten by role, since the split, the R-flux/W paper, the two contraction-classification papers, and the four-level-shape paper are prerequisites this document needed to make the chain self-contained rather than documents Foundation cites directly. Dependencies here are traced through every citation used inside each paper's proofs, not only its stated citation list; the two methods differ, and only the first is reliable — a stated list can silently omit a load-bearing in-proof citation.

A note on notation, assembled from ten source papers each with its own local conventions: four symbols carry more than one meaning across this document, resolved here rather than left to context. μ denotes the order-three automorphism cycling the three conjugate splits of §1, throughout; the second-contraction parameter of §§3–5, called μ in the source paper, is written $s \in [0, 1]$ here instead to keep the two apart. D denotes the squeeze generator of §§4 and 7 ($D = \frac{1}{2}\sum_a (X_a P_a + P_a X_a)$, later $-i\partial_u$); the one-particle test space of §§8–13, called D in the source paper, is written $\mathcal{D}$ here instead. W denotes the central structural element $W = \hbar \cdot 1$ throughout, unchanged, since this is the symbol every other document in this project already uses for it; the two-point kernel of §8, which the source paper also calls W, is written $\mathcal{K}(f, g)$ here instead, and the Weyl unitary operators $W(f)$ keep their standard name, distinguished from the central element by always carrying an argument. α denotes the physical dynamics (§§8, 11), kept apart from the split involution τ of §1, which the source papers sometimes use for both — a companion document in this project (*The Permission Slip*) writes this same dynamics as S_t instead, reusing its own squeeze-flow name; the two choices are a difference of local convention between companion documents, not a further collision within either one. λ carries two standard, unrelated meanings left as found — the dilation parameter of §4, and the Connes subtype label of §§10 and 13 (III_λ) — since both are sufficiently frozen conventions in their own literatures that renaming either would trade one confusion for another; context disambiguates in every instance.

[LEDGER] Registry items this document depends on: REG-1 (injectivity, second-reader confirmation), REG-3 ([OPEN 1]/[OPEN 3] fusion question), REG-4 (the μ/χ question), REG-5 (locator debts for companion papers and cited theorems), REG-8 (the $\mathrm{III}_1/\mathrm{III}_\lambda$ fork, this chain's own first genuinely falsifiable prediction-shape, §13). Local to this document: which triple the second contraction identifies as physical space and why, given that Orbit A's route and Orbit B's route provably reach the identical Heisenberg algebra while disagreeing on which of the two canonically-paired triples plays X — a structural fact, verified directly; the physical question of which identification is correct remains open, with the founding R-flux literature's own explicit construction — not merely a commutator-sign convention — supporting Orbit B's identification, though still not deriving it (§3); whether §9's channels use the raw or orthonormalized angular basis, a one-sentence convention this document argues for but does not fix (§9); the reconciliation with the thermal-state classification for conformal nets on nets retaining charge and zero-mode structure (§12).

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